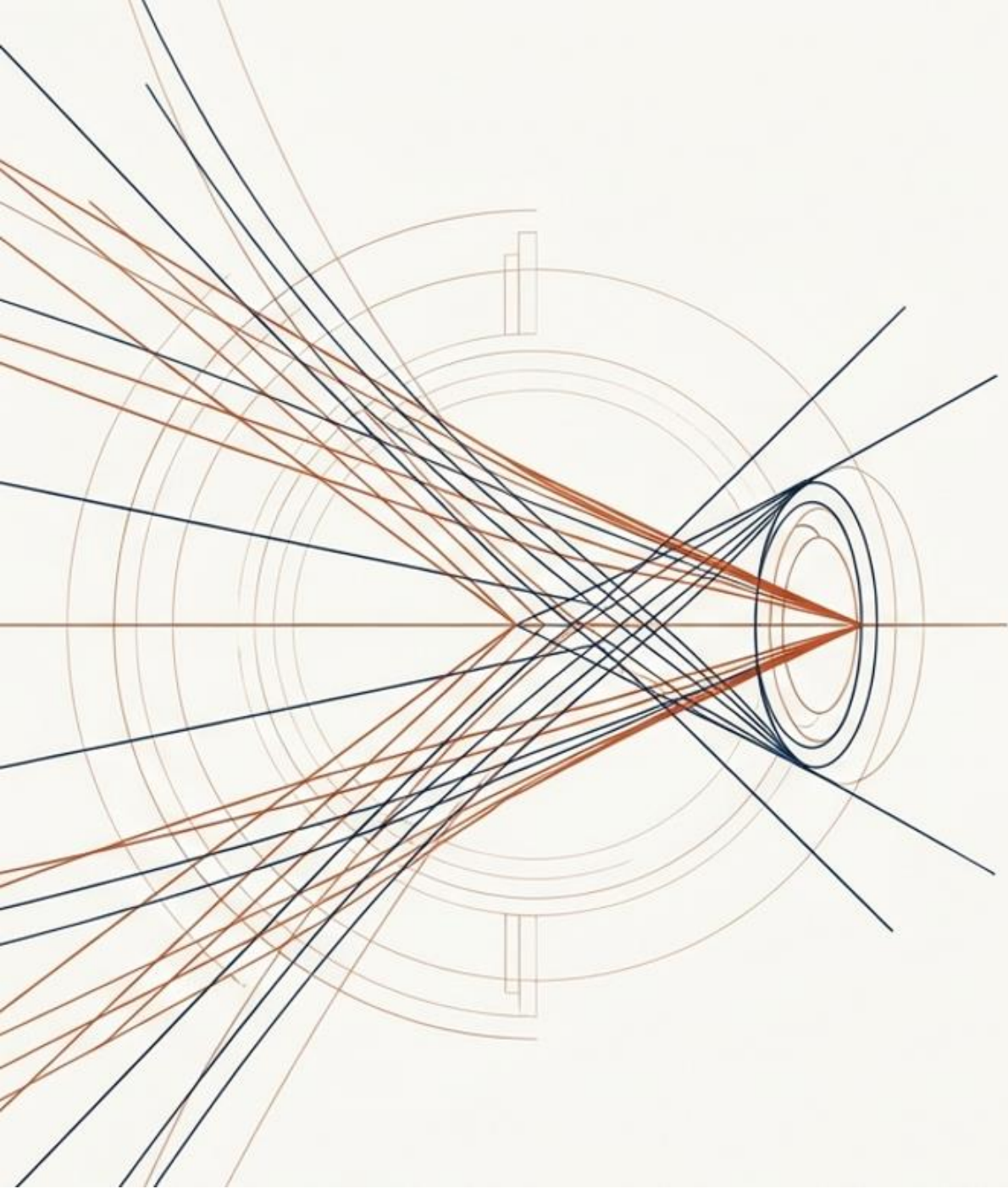


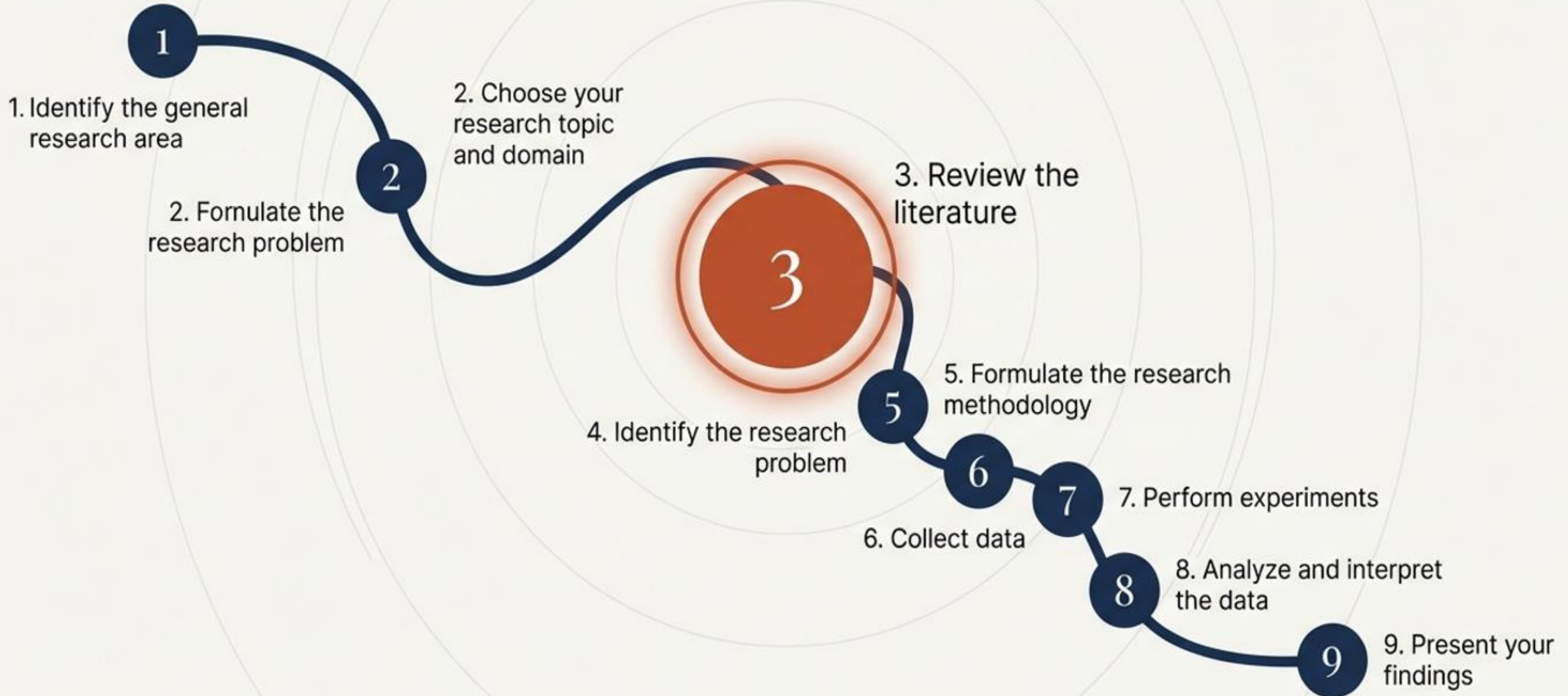


Mastering the Research Process: From Broad Domains to Critical Critique

Developing Essential Skills for
Research Students

Based on ARI-304 Research Methods
Prof. Dr. Faheem Akhtar Rajput

The Overarching Research Lifecycle



Finding the Gap: Immersion Precedes Discovery

A good research problem (e.g., "Predicting breast cancer from mammogram images") is never found in a vacuum. The prerequisite is extensive background knowledge."



Read iconic, foundational papers



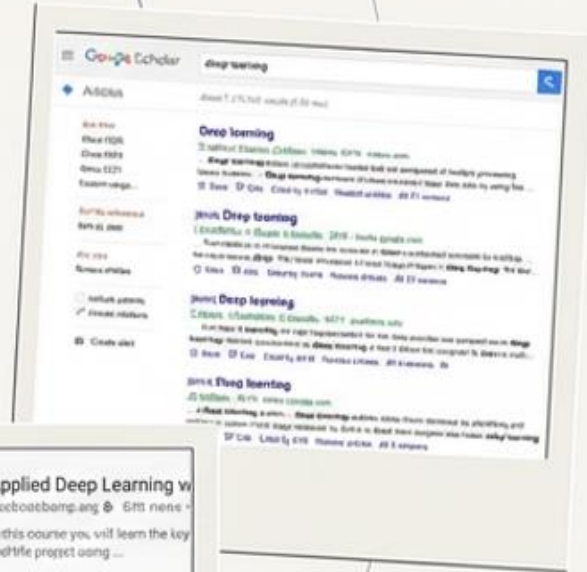
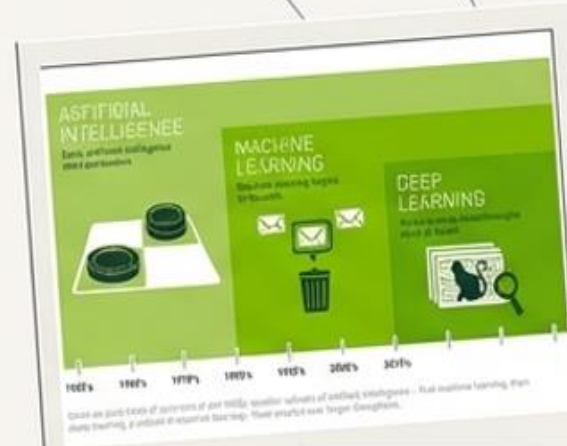
Read comprehensive book chapters



Watch full video lectures and courses



Listen to expert talks and industry reports



The Engine of Discovery: The Literature Review

“An account of what has been published by accredited scholars, acting as the foundation and support for new insight.”

Information Seeking

Efficiently scanning the literature.

Critical Appraisal

Identifying valid, unbiased studies.

Synthesis

Highlighting strengths, weaknesses, and limitations of existing work.

Justification

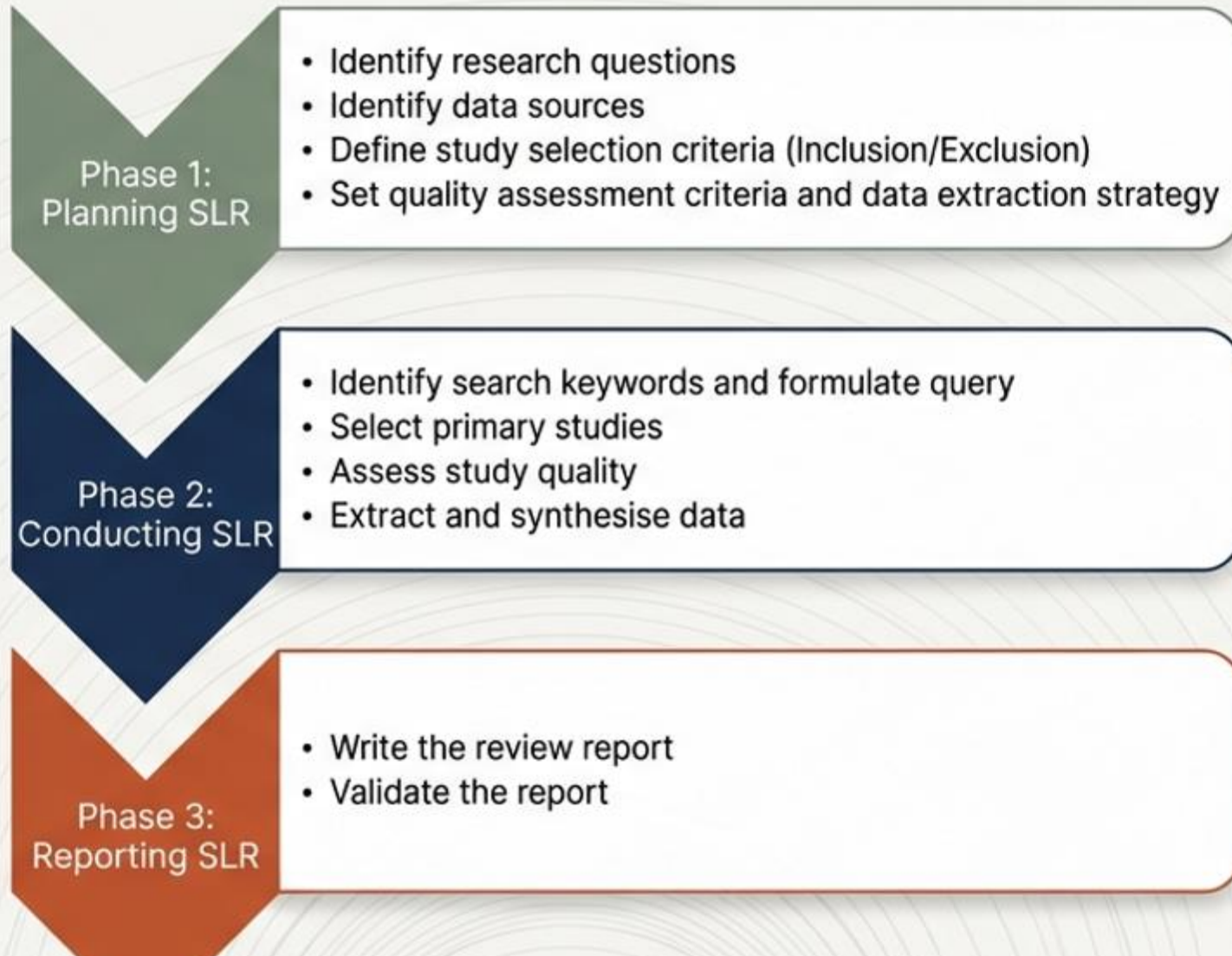
Shaping the research problem and defining its relationship to the existing body of knowledge.



Traditional vs. Systematic Literature Review (SLR)

	Systematic Literature Review (SLR)	Traditional Literature Review
Question:	Focuses on a single, specific question (often PICO based).	Broad overview; not necessarily focused on a single question.
Protocol:	Includes a peer-reviewed plan or protocol.	No protocol included.
Inclusion Criteria:	Explicitly states criteria before conducting the review.	Criteria not explicitly stated.
Search Strategy:	Comprehensive and systematic.	Not explicitly stated; may not be comprehensive.
Evaluation:	Comprehensive evaluation of study quality.	May or may not include quality evaluation.

The 3 Phases of a Systematic Literature Review



“Reading without critical evaluation is like building on sand.”



Not all published papers are equally valid.



Peer review reduces errors but doesn't eliminate them.



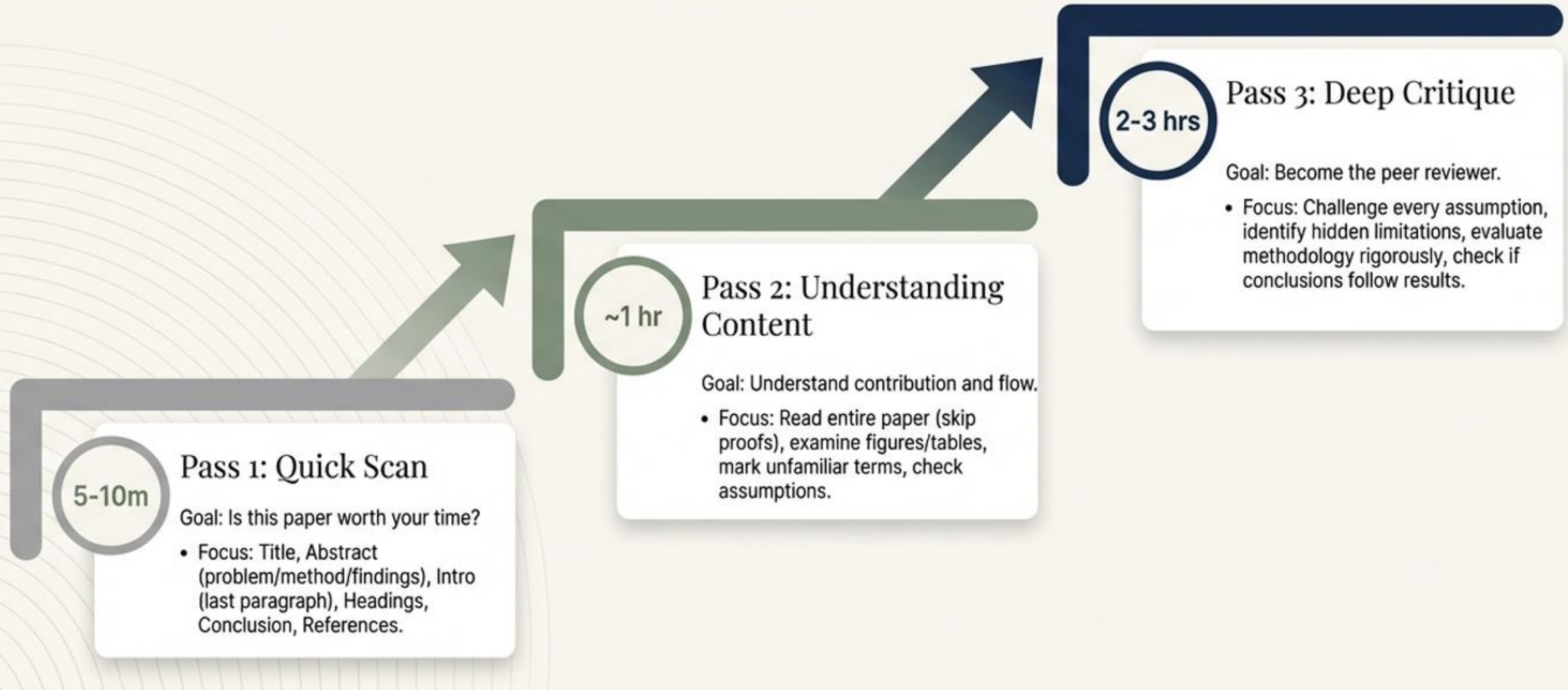
Your research builds on existing work—it must be solid.



Identifying research gaps explicitly requires understanding the limitations of others.

The Three-Pass Approach to Reading

Based on S. Keshav's Methodology



The CRAAP Test for Research Papers

C

Currency

Is the research recent? Are the references up-to-date?

R

Relevance

Does it directly address your specific research question?

A

Authority

Are the authors qualified? Is the journal/conference credible (impact factor, indexing)?

A

Accuracy

Is the methodology sound? Are the claims actually supported?

P

Purpose

Why was this research done? Is there any underlying bias?

Deconstructing the Setup: Introduction & Literature Review

Evaluating the Introduction

✓ Introduction Checklist

Clear problem statement, well-defined gap, explicit contribution, motivation provided.

⚠ Introduction Red Flags

Vague problem, no gap identified, overstated importance.

Evaluating the Literature Review

✓ Literature Review Checklist

Comprehensive yet critical (not merely descriptive), includes landmark papers, includes recent papers (last 3-5 years), establishes context for this research.

⚠ Literature Review Red Flags

Old references, missing key papers, no critical analysis of previous limitations.

Deconstructing the Core: Methodology & Results

Evaluating Methodology

Empirical Research:

- Is sample size adequate/justified?
- Are data collection methods reliable?
- Could the study be replicated?

Computational Research:

- Is the dataset described (size/source)?
- Are algorithms specified with parameters?
- Are evaluation metrics appropriate?
- Is there comparison
- Is there comparison with baselines?

Evaluating Results

? Questions to Ask:

- Do the results actually answer the research questions?
- Are statistical tests appropriate?
- Are negative/unexpected results reported?

! Red Flags:

- No error bars
- Cherry-picking results
- Over-interpretation of data

Deconstructing the Synthesis: Discussion & Conclusion

Evaluating the Wrap-Up:

- ✓ Results must be interpreted in the context of the wider literature.
- ✓ Theoretical and practical implications are clearly discussed.
- ✓ Conclusions are strictly supported by the actual results (no leaps of logic).
- ✓ Future work suggestions are provided based on findings.



The Honesty Check

Limitations must be honestly acknowledged.



Red Flags

- Claims not supported by data.
- Limitations intentionally hidden.
- Overgeneralisation of the findings.

Spotting the Pitfalls: A Warning Signs Gallery



Sampling Bias

Relying on convenience samples or sample sizes that are too small.



Confirmation Bias

Ignoring contradictory evidence to fit a narrative.



P-Hacking

Fishing through data simply to find statistically significant results.



Overclaiming

Drawing conclusions that stretch far beyond what the data actually supports.



Missing Baselines

Failing to compare the new work with existing, established methods.



Cherry-Picking

Selectively reporting only the favourable results while hiding the negative ones.



Unclear Methodology

A study designed in a way that makes replication impossible.

The Synthesis Dashboard: Extracting the Value

The Four-Question Framework

If you can answer these, you've understood the paper:

1. What did the authors try to do?
(Research question)
2. How did they try to do it?
(Methodology)
3. What did they find? (Results)
4. Do I believe the findings? (Critical judgment)

The Note-Taking Template

Citation [Full reference]

Research Problem

Methodology

Key Findings

Strengths / Weaknesses & Limitations

 /

Questions for Authors

Relevance to My Research

Key References to Follow

Building the Muscle: Making Critique a Habit

Critical reading is not a one-off task; it is a skill that requires regular practice and refinement. Focus on methodology, identify limitations to find your own research opportunities, and question everything.

Read with a peer

Discuss, debate, and defend your critiques.

Write short reviews

Practice articulating your critical thoughts in writing.

Compare with rebuttals

Read authors' responses to criticisms to see both sides.

Join journal clubs

Learn from the diverse perspectives of other researchers.

Mock Peer Review

Review papers for journals to simulate the actual peer-review process.